

CHE 311 HW # 8: 7A.1, .2, .4; 7B.1

7A.1 Pressure rise in a sudden enlargement

rate = 450 US gal/min = 0.281 m³/s

$d_5 = 5 \text{ in.}$ $d_L = 9 \text{ in.}$

a) What is the pressure rise in pounds per in² if $\mu = 63 \text{ lbm/ft}^3$?

b) IS flow is smaller, pipe laminar or turbulent? NOT enough info given, need Reynolds #.

a) Using Eqn. 7.6-4,

$$p_2 - p_1 = \rho v_2^2 \left(\frac{1}{\beta} - 1 \right)$$

and $\beta = \frac{S_1}{S_2} = \left(\frac{d_L}{d_5} \right)^2 = \left(\frac{5 \text{ in}}{9 \text{ in}} \right)^2 = .3086419$

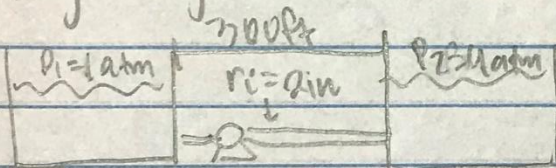
Re 1200 laminar

Re > 4000 turbulent

and $v_2 = \frac{Q}{S_2} = \frac{450 \text{ gal/min}}{\pi (2.5 \text{ ft})^2} = 7.3529 \frac{\text{ft}}{\text{s}}$

$$\Delta p = 63 \frac{\text{lbm}}{\text{ft}^3} \cdot (7.3529 \frac{\text{ft}}{\text{s}})^2 \left(\frac{1}{.3086419} - 1 \right) = 7610 \frac{\text{lbm}}{\text{ft}} = 1.64 \text{ psi}$$

7A.2 Pumping a hydrochloric acid solution



$\mu = 62.4 \text{ lbm/ft}^3$

$M = 1 \text{ ap}$

$Re = 7.11 \times 10^4 \rightarrow \text{Turbulent}$

$\langle v \rangle = 2.30 \text{ ft/s}$

What is the power delivered to the pump?

Using Eqn 7.5-10

$$\dot{W}_m = \underbrace{p_2 - p_1}_{(1)} + \underbrace{\frac{1}{2} v^2 \frac{L}{R_h} f}_{(2)} + \underbrace{\frac{1}{2} v^2}_{(3)}$$

$f = .0048$ from Eq 6.2-12

(1) $p_2 - p_1 = (4 - 1) \text{ atm} \cdot 6.8087 \times 10^{-4} \frac{\text{lbm}}{\text{ft}^3} \cdot \frac{1}{\text{atm}} = 3273 \frac{\text{ft}^2}{\text{s}^2}$

(2) $\frac{1}{2} v^2 \frac{L}{R_h} f = \frac{1}{2} (2.30 \frac{\text{ft}}{\text{s}})^2 \frac{300 \text{ ft}}{0.5/12 \text{ ft}} \cdot (.0048) = 92 \frac{\text{ft}^2}{\text{s}^2}$

(3) $\frac{1}{2} v^2 = \frac{1}{2} (2.30 \frac{\text{ft}}{\text{s}})^2 = 2.645 \frac{\text{ft}^2}{\text{s}^2}$

$\dot{W}_m = (3273 + 92 + 2.645) \text{ ft}^2/\text{s}^2 = 3367.645 \text{ ft}^2/\text{s}^2$

Note

cont.

Determine mass flow rate,

$$W = (\pi R^2) \rho \langle v \rangle = \pi \frac{4}{144} \text{ ft}^2 \left(62.4 \frac{\text{lbm}}{\text{ft}^3} \right) \left(2.30 \frac{\text{ft}}{\text{s}} \right) = 12.5 \frac{\text{lbm}}{\text{s}}$$

Power is $\dot{W}_m$, $W = 3368 \frac{\text{ft}^2}{\text{s}^2} \cdot 12.5 \frac{\text{lbm}}{\text{s}}$

$$= 4.22 \times 10^4 \frac{\text{ft} \cdot \text{lbm}}{\text{s}} \cdot \frac{3600 \text{ s}}{\text{hr}} \cdot \frac{1.5 (98 \times 10^{-8} \text{ hp} \cdot \text{hr})}{\text{ft} \cdot \text{lbm}} \quad \boxed{2.4 \text{ hp} = 1.8 \text{ kW}}$$

7A.4 Incompressible Flow in an Annulus

H₂O @ 60°F delivered from pump through annular conduit $L = 20.3 \text{ ft}$, rate = 241 US gal/min

$r_i = 3 \text{ in}$ $r_o = 7 \text{ in}$ Inlet is 5 ft lower than outlet

Determine the power input required from pump by using mean hydraulic empiricism. Pressures @ pump inlet = outlet

Using Eqn 6.2-16, mean hydraulic radius,

$$R_h = \frac{S}{2} = \frac{\pi (R_o^2 - R_i^2)}{2\pi (R_o + R_i)} = \frac{1}{2} (R_o - R_i)$$

Mass Flow Rate,

$$W = \frac{241}{60} \frac{\text{US gal}}{\text{s}} \cdot \frac{231}{1728} \frac{\text{ft}^3}{\text{gal}} \cdot 62.4 \frac{\text{lbm}}{\text{ft}^3} = 33.5 \frac{\text{lbm}}{\text{s}}$$

Average Flow Velocity,

$$\langle v \rangle = \frac{W}{\rho S} = \frac{W/\rho}{\pi (R_o^2 - R_i^2)} = \frac{33.5 \frac{\text{lbm}}{\text{s}} \cdot 1 \text{ ft}^3}{62.4 \frac{\text{lbm}}{\text{ft}^3} \pi (7/12)^2 - (3/12)^2 \text{ ft}^2} = 0.015 \frac{\text{ft}}{\text{s}}$$

Reynolds #,

$$\begin{aligned} Re &= \frac{\mu R_h \langle v \rangle \rho}{\mu} = \frac{2(R_o - R_i) W}{\pi (R_o^2 - R_i^2) \mu} = \frac{2W}{\pi (R_o + R_i) \mu} = \frac{2(33.5 \frac{\text{lbm}}{\text{s}})}{\pi (7+3/12) \text{ ft} (1.14 \cdot 10^{-6} \text{ lbm})} \\ &= 3.34 \times 10^4 \rightarrow \text{turbulent flow} \\ f &= .0059 \end{aligned}$$

Use Eqn 7.5-10 to find work

$$\dot{W} = g(h_2 - h_1) + \frac{1}{2} \langle v \rangle^2 \frac{L}{R_h} f = g(h_2 - h_1) + \langle v \rangle^2 \frac{L}{(R_o - R_i) f}$$

$$= 32 \frac{\text{ft}}{\text{s}^2} \cdot 5 \text{ ft} + (0.015 \frac{\text{ft}}{\text{s}})^2 \left(\frac{20.3}{(7-3/12)} \right) (.0059) = 161 + .175 = 161.1 \frac{\text{ft}^2}{\text{s}^2}$$

cont

Power is $w \cdot \dot{W}$ so

$$\frac{33.5 \text{ lbm}}{s} \cdot \frac{161.1 \text{ ft}^2}{s^2} = \frac{5.4 \times 10^3 \text{ ft}^3 \text{ lbm}}{s} = \frac{1.9 \times 10^7 \text{ ft}^3 \text{ lbm}}{\text{hr}}$$

$$= 0.31 \text{ hp} = 0.23 \text{ kW}$$

7B.1 Velocity averages from the $1/7$ Power Law

Evaluate ratios in 7A.7 according to vel. distribution in Eqn 5.1-4

a) Average velocity

$$\frac{\langle \bar{v}_z \rangle}{v_{\max}} = \frac{\int_0^{2\pi} \int_0^R [1 - (r/R)]^{1/7} r dr d\theta}{\int_0^{2\pi} \int_0^R r dr d\theta} = 2 \int_0^1 (1 - \phi)^{1/7} \phi d\phi$$

$$= 2 \int_0^1 x (1 - x^{7/2})^{1/7} dx = 14 (1/8 - 1/15) = \boxed{49/60} \cdot$$

b) Average of the square velocity

$$\frac{\langle \bar{v}_z^2 \rangle}{v_{\max}^2} = \frac{\int_0^{2\pi} \int_0^R [1 - (r/R)]^{2/7} r dr d\theta}{\int_0^{2\pi} \int_0^R r dr d\theta} = 2 \int_0^1 (1 - \phi)^{2/7} \phi d\phi$$

$$= 2 \int_0^1 x (1 - x^{7/2})^{2/7} dx = 7 \left(\frac{2}{7} - \frac{1}{8} \right) = \boxed{\frac{49}{72}} \cdot$$

$$\text{so that } \frac{\langle \bar{v}_z^2 \rangle}{\langle \bar{v}_z \rangle^2} = \frac{49}{72} \left(\frac{60}{49} \right)^2 = \frac{50}{49} \checkmark$$

c) Average of the cube of the velocity

$$\frac{\langle \bar{v}_z^3 \rangle}{v_{\max}^3} = \frac{\int_0^{2\pi} \int_0^R [1 - (r/R)]^{3/7} r dr d\theta}{\int_0^{2\pi} \int_0^R r dr d\theta} = 2 \int_0^1 (1 - \phi)^{3/7} \phi d\phi$$

$$= 2 \int_0^1 x (1 - x^{7/2})^{3/7} dx = \frac{14}{3} \left(\frac{3}{10} - \frac{3}{17} \right) = \boxed{\frac{49}{85}} \cdot$$

$$\text{so that } \frac{\langle \bar{v}_z^3 \rangle}{\langle \bar{v}_z \rangle^3} = \frac{49}{85} \left(\frac{60}{49} \right)^3 = \frac{43200}{24017} \checkmark$$